2.2.3 Orthonormal vectors and matrices

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Week 02 due Jan 18, 2025 00:01 IST

Video

Okay, now for historical reasons when we start

talking about such matrices we tend to use the letter q for the columns. So I'm

going to change the u here to a q and say okay we have n mutually

orthonormal vectors q_0 through q_n-1, all in C^m. And we know that they

have this property right here. Now let's take those vectors and let's make them

columns of a matrix. What happens if you look at Q Hermitian transpose times Q?

5:36 / 5:36 2.0x

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Reading assignment

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Read Unit 2.2.3 of the notes. [LINK]

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Homework 2.2.3.1

1.0/1.0 point (graded)

Let $u \neq 0 \in \mathbb{C}^m$.

$u \;/\; \|u\|_2$ has unit length.

ALWAYS ✔ Answer: ALWAYS

Solution:

Now prove it!

For complete solution, see [LINK].

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Homework 2.2.3.2

1.0/1.0 point (graded)

Let $Q \in \mathbb{C}^{m \times n}$ (with $n \leq m$). Partition $Q = \left(\begin{array}{c|c|c|c} q_0 & q_1 & \cdots & q_{n-1} \end{array} \right)$.

Q is an orthonormal matrix if and only if $q_0, q_1, \ldots, q_{n-1}$ are mutually orthonormal.

TRUE ✔ Answer: TRUE

Solution:

Now prove it!

For complete solution, see [LINK].

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Homework 2.2.3.3

1/1 point (graded)

Let $Q \in \mathbb{C}^{m \times n}$ and $Q^H Q = I$.

$QQ^H = I$.

SOMETIMES ✔ Answer: SOMETIMES

Solution:

Now explain why!

Now explain why:

For complete solution, see [LINK].

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